MH1200 Linear Algebra I.

Problem set #10.

SOLUTIONS

Problem 1:

Consider the following list of vectors in $\mathbb{R}^4$ depending on a constant a:

$$\mathbf{v_1} = (1, a, 0, -2)\ , \quad \mathbf{v_2} = (0, 1, -a, 2a)\ , \quad \mathbf{v_3} = (1, 3, 0, -2)$$

where a is a real number. For what values of a is this collection of vectors linearly independent?

Solution

Put the vectors as columns in a matrix and apply Gaussian elimination to reduce it to row-echelon form. From the lectures we know that the vectors are linearly independent if and only if the number of non-zero rows in the row-echelon matrix is the same as the number of columns, i.e. 3 in this case.

When doing this it is always a good idea to chose the order of the vectors such that the unknown constant a is as far as possible to the right in the matrix. (This usually makes the Gaussian elimination less complicated.) In this case we chose the order $\mathbf{v_3}, \mathbf{v_1}, \mathbf{v_2}$ when putting the vectors as columns in the matrix. Then the matrix and Gaussian elimination are as follows:

$$\begin{pmatrix} 1 & 1 & 0 \\ 3 & a & 1 \\ 0 & 0 & -a \\ -2 & -2 & 2a \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & a-3 & 1 \\ 0 & 0 & -a \\ 0 & 0 & 2a \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & a-3 & 1 \\ 0 & 0 & -a \\ 0 & 0 & 0 \end{pmatrix} \equiv R.$$

If $a \neq 3$ and $a \neq 0$ then the final matrix R is in row-echelon form and has 3 non-zero rows so the vectors are linearly independent in this case. If $a = 0$ there are only 2 non-zero rows so the vectors are linearly dependent. If $a = 3$ we need to further reduce R to get it into row-echelon form:

$$R = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Again there are only 2 non-zero rows in the final row-echelon matrix, so the vectors are also linearly dependent in this case.

To summarize, the vectors are linearly independent for all values of a except for $a = 3$ and $a = 0$.

$\square$

Problem 2: (From final exam 2018.)

Let a be a real constant, and consider the following list of vectors in $\mathbb{R}^4$:

$$S = \left\{(1,0,1,a^2),\ (0,1,-3,a),\ (1,1,-2,0)\right\}.$$

(i) For what values of the constant a is S linearly independent?

(ii) For what values of the constant a is it true that $\operatorname{span}(S) = \mathbb{R}^4$?

Solution

We can just follow the algorithm we learnt in lectures. We start by making a matrix whose columns are the given vectors. It will be convenient to reorder the vectors from last to first and then we write down:

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -2 & -3 & 1 \\ 0 & a & a^2 \end{bmatrix}.$$

Now we do Gaussian elimination:

$$\to \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -3 & 3 \\ 0 & a & a^2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & a^2+a \\ 0 & 0 & 0 \end{bmatrix}$$

If $a = 0$ or $a - 1$ then this matrix becomes

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

In this case the third column of this matrix in row echelon form doesn't contain a leading entry, so the corresponding vector in the original list is a linear combination of the other two vectors.

But in all other cases $a \neq 0, -1$ the row echelon form is:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & \neq 0 \\ 0 & 0 & 0 \end{bmatrix}$$

For part (ii): There is no value of a that will result in $\operatorname{span}(S) = \mathbb{R}^4$. There are many ways to see this but the basic issue is that you need at least 4 different vectors to span $\mathbb{R}^4$ which has dimension 4. One concrete way to see this is that we learnt an algorithm that says the vectors will span $\mathbb{R}^4$ if and only if when you build a mtrix with the vectors as the columns, then a row echelon form has no zero rows. In the case that you only have 3 vectors each with 4 entries, your starting matrix is 4×3, and Gaussian elimination will definitely leave you with a non-zero row.

$\square$

Problem 3:

Express each one of the vectors on the following list of vectors in $\mathbb{R}^3$ as a linear combination of the other vectors on the list.

$$(1, 0, -1), (-1, 2, 3), (0, 3, 0), (1, -1, 1).$$

Solution

We need to understand the solutions to the equation

$$c_1(1, 0, -1) + c_2(-1, 2, 3) + c_3(0, 3, 0) + c_4(1, -1, 1) = (0, 0, 0).$$

The corresponding homogeneous system is

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & 3 & -1 & 0 \\ -1 & 3 & 0 & 1 & 0 \end{array}\right].$$

Now we solve this system:

$$\to \left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & 3 & -1 & 0 \\ 0 & 2 & 0 & 2 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & 3 & -1 & 0 \\ 0 & 0 & -3 & 3 & 0 \end{array}\right]$$

Using back-substitution we can write down the general solution. We set c_4 to a free parameter $c_4 = s$ and solve. The last equation gives:

$$c_3 = c_4 = s.$$

The second last equation gives:

$$2c_2 + 3s - s = 0 \quad \Rightarrow \quad c_2 = -s.$$

Finally the first equation gives:

$$c_1 = c_2 - c_4 = -s - s = -2s.$$

So we deduce that for every choice of $s \in \mathbb{R}$ we get:

$$-2s(1, 0, -1) - s(-1, 2, 3) + s(0, 3, 0) + s(1, -1, 1) = (0, 0, 0).$$

In particular if we set $s = 1$ this becomes:

$$-2(1, 0, -1) - (-1, 2, 3) + (0, 3, 0) + (1, -1, 1) = (0, 0, 0).$$

We can easily rearrange this in 4 different ways to express each vector in terms of the other vectors:

- $(1, 0, -1) = -\frac{1}{2}(-1, 2, 3) + \frac{1}{2}(0, 3, 0) + \frac{1}{2}(1, -1, 1)$.
- $(-1, 2, 3) = -2(1, 0, -1) + (0, 3, 0) + (1, -1, 1)$.
- $(0, 3, 0) = 2(1, 0, -1) + (-1, 2, 3) - (1, -1, 1)$.

- $(1,-1,1) = 2(1,0,-1) + (-1,2,3) - (0,3,3)$.

$\square$

Problem 4: (From final exam 2018.)

Let $T = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be some linearly independent list of vectors in $\mathbb{R}^4$. Is either of the lists

(i) $T' = \{\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_2 - \mathbf{v}_3, \mathbf{v}_1 - \mathbf{v}_3\}$

(ii) $T'' = \{\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_2 - \mathbf{v}_3, \mathbf{v}_1 + \mathbf{v}_3\}$

also linearly independent? Justify your answer.

Solution to (i).

We can easily see straight away that the list

$$\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_2 - \mathbf{v}_3, \mathbf{v}_1 - \mathbf{v}_3$$

is not linearly independent because

$$(\mathbf{v}_1 - \mathbf{v}_3) = (\mathbf{v}_1 - \mathbf{v}_2) + (\mathbf{v}_2 - \mathbf{v}_3).$$

Solution to (ii).

What we know: Because the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly independent, the only solution to the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = 0$$

is $c_1 = c_2 = c_3 = 0$.

We will use this to understand the solutions to the equation:

$$d_1(\mathbf{v}_1 - \mathbf{v}_2) + d_2(\mathbf{v}_2 - \mathbf{v}_3) + d_3(\mathbf{v}_1 + \mathbf{v}_3) = 0.$$

If we rearrange this equation to group the terms v_i then it becomes:

$$(d_1 + d_3)\mathbf{v}_1 + (d_2 - d_1)\mathbf{v}_2 + (d_3 - d_2)\mathbf{v}_3 = 0.$$

Using the fact that the $\mathbf{v}_i$ are linearly independent each of these coefficients must be zero. This implies the system of linear equations:

$$\begin{cases} d_1 + d_3 = 0 \\ -d_1 + d_2 = 0 \\ d_3 - d_2 = 0 \end{cases}$$

Solving this using our standard techniques we get:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right].$$

Thus the unique solution is $d_1 = d_2 = d_3 = 0$.

This implies that the list T' is indeed linearly independent.

$\square$

Problem 5

Consider a linearly independent list of m vectors in some $\mathbb{R}^n$

$$\mathbf{v}_1, \ldots, \mathbf{v}_m \in \mathbb{R}^n.$$

Let $\mathbf{w} \in \mathbb{R}^{\mathbf{n}}$ be a vector with the property that when it is added to this list

$$\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{w}$$

then the resulting list is linearly dependent. Prove that: $\mathbf{w} \in \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$.

Solution

Because the list of vectors

$$\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{w}$$

is linearly dependent, we know that there exist scalars $c_1, \ldots, c_m, d$ such that

$$c_1\mathbf{v}_1 + \ldots + c_m\mathbf{v}_m + d\mathbf{w} = \mathbf{0}$$

where not all of these coefficients are zero.

Next note that d cannot equal 0 because that would imply an equation

$$c_1\mathbf{v}_1 + \ldots + c_m\mathbf{v}_m = \mathbf{0}$$

with not all $c_i = 0$. This is impossible because we know that the list

$$\mathbf{v}_1, \ldots, \mathbf{v}_m \in \mathbb{R}^n$$

is linearly independent.

Thus because $d \neq 0$ we can rearrange this equation as follows:

$$\mathbf{w} = -\frac{c_1}{d}\mathbf{v}_1 + \ldots - \frac{c_m}{d}\mathbf{v}_m$$

By definition this shows that $\mathbf{w} \in \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$.

$\square$

Problem 6: (From final exam 2020.)

(a) Give an example of a 3×3 matrix such that any two of the three rows are linearly independent but such that the set of all three rows is linearly dependent.

(b) Show that an $n \times n$-matrix $\mathbf{A}$ has determinant equal to zero if and only if the rows of $\mathbf{A}$ are linearly dependent.

You may use standard facts proved in lectures.

Solution to Part (a).

It is enough, for example, to write down any 3×3 matrix whose first two rows are linearly independent (non-zero and non-parallel is enough) and then to make the third row the sum of the first two rows. Such as:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}.$$

(This is because if the pair $\vec{r}_1$ and $\vec{r}_2$ is linearly independent then so it the pair $\vec{r}_1$ and $\vec{r}_1 + \vec{r}_2$ and the pair $\vec{r}_2$ and $\vec{r}_1 + \vec{r}_2$.)

Solution to Part (b).

We'll prove the two directions separately.

First we prove that (rows are linearly dependent) implies (determinant is zero).

If the rows $\left\{\vec{R}_i\right\}$ are linearly dependent then one of the rows can be expressed as a linear combination of the other rows:

$$\vec{R}_i = c_1\vec{R}_1 + \ldots + \widehat{c_i\vec{R}_i} + \ldots + c_n\vec{R}_n.$$

(Here the notation indicates the i-th term is omitted.) Then the determinant can be expanded using the property that determinants are linear functions of rows:

$$\begin{vmatrix} \leftarrow & \vec{R}_1 & \rightarrow \\ & \vdots & \\ \leftarrow & \vec{R}_i & \rightarrow \\ & \vdots & \\ \leftarrow & \vec{R}_n & \rightarrow \end{vmatrix} = \sum_{k=1,k\neq i}^{n} c_k \begin{vmatrix} \leftarrow & \vec{R}_1 & \rightarrow \\ & \vdots & \\ \leftarrow & \vec{R}_k & \rightarrow \\ & \vdots & \\ \leftarrow & \vec{R}_n & \rightarrow \end{vmatrix}.$$

Every determinant in this sum has a repeated row, so the expression is zero.

Next we prove that: (determinant is zero) implies (rows are linearly dependent).

If the determinant is zero, then the determinant of $\mathbf{A}^T$ is also zero. Thus $\mathbf{A}^T$ is not invertible, by the theorem that determinant detects invertibility. Then by the characterization of invertibility, we know that there is a non-zero vector $\vec{x}$ such that

$$\mathbf{A}^T\vec{x} = \vec{0}.$$

This gives us an equation:

$$x_1\vec{R}_1 + \ldots + x_n\vec{R}_n = \vec{0}$$

where $\vec{R}_i$ denote the rows of $\mathbf{A}$ and where not all of the x_i are zero.

Thus the rows of $\mathbf{A}$ are linearly dependent. $\square$

Problem 7. (From final exam 2021.)

(a) Is the following list of 3 vectors in $\mathbb{R}^4$ linearly dependent or independent?

$$\vec{w}_1 = (1, 4, 2, -3)\,,\ \vec{w}_2 = (7, 10, -4, -1)\,,\ \vec{w}_3 = (-2, 1, 5, -4).$$

(b) Let $\vec{u}$ and $\vec{v}$ be a linearly independent pair of vectors in some $\mathbb{R}^n$. Use the definition of linear independence to carefully prove that if $a, b, c, d \in \mathbb{R}$ satisfy $ad - bc \neq 0$ then the vectors

$$\vec{x} = a\vec{u} + b\vec{v}\ ,\quad \vec{y} = c\vec{u} + d\vec{v}$$

are also linearly independent.

Solution to (a).

This is just an application of the algorithm to test if vectors are linearly independent. First make the vectors the columns of a matrix:

$$\begin{bmatrix} 1 & 7 & -2 \\ 4 & 10 & 1 \\ 2 & -4 & 5 \\ -3 & -1 & -4 \end{bmatrix}.$$

The we do Gaussian elimination:

$$\to \begin{bmatrix} 1 & 7 & -2 \\ 0 & -18 & 9 \\ 0 & -18 & 9 \\ 0 & 20 & -10 \end{bmatrix} \to\to \begin{bmatrix} 1 & 7 & -2 \\ 0 & -18 & 9 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The third column doesn't have a leading entry, so the third vector given at the start $\vec{w}_3$ may be expressed as a linear combination of the other two vectors $\vec{w}_1$ and $\vec{w}_2$.

Solution to (b).

Assume that r_1 and r_2 are real numbers such that

$$r_1(a\vec{u} + b\vec{v}) + r_2(c\vec{u} + d\vec{v}) = \vec{0}.$$

If follows that

$$(r_1 a + r_2 c)\vec{u} + (r_1 b + r_2 d)\vec{v} = \vec{0}.$$

Because $\vec{u}$ and $\vec{v}$ are linearly independent it follows that

$$\begin{aligned} ar_1 + cr_2 &= 0 \\ br_1 + dr_2 &= 0 \end{aligned}$$

Taking d times the first equation minus c times the second we deduce $(ad - bc)r_1 = 0$. Thus because $ad - bc = 0$ we deduce $r_1 = 0$. Similarly we deduce $r_2 = 0$.

Thus $\vec{x}$ and $\vec{y}$ are linearly independent.

□

Problem 8. (From final exam 2021.)

(a) Calculate the adjoint matrix of the following matrix. Show full working.

$$\mathbf{A} = \begin{bmatrix} -3 & 2 & -5 \\ -1 & 0 & -2 \\ 3 & -4 & 1 \end{bmatrix}$$

(b) Is the following statement true or false? Justify your answer.

Statement: If $\mathbf{M}$ is a non-zero matrix satisfying $\det(\mathbf{M}) = 0$ then the list of rows of the adjoint matrix of $\mathbf{M}$ is linearly dependent.

Solution to (a).

This is a straightforward computation from the definition of adjoint. We take the transpose of the matrix of cofactors. Here:

$$\operatorname{adj}(\mathbf{A}) = \begin{bmatrix} -8 & (-1)5 & 4 \\ (-1)(-18) & 12 & (-1)6 \\ (-4) & (-1)1 & 2 \end{bmatrix}^T = \begin{bmatrix} -8 & 18 & -4 \\ -5 & 12 & -1 \\ 4 & -6 & 2 \end{bmatrix}.$$

Solution to (b).

Assume $\mathbf{M}$ is an $n \times n$ matrix. If $\det(\mathbf{M}) = 0$ then the fundamental property of the adjoint $\mathbf{M}\operatorname{adj}(\mathbf{M}) = \det(\mathbf{M})\mathbf{I}_n$ becomes:

$$\mathbf{M}\operatorname{adj}(\mathbf{M}) = \mathbf{0}.$$

The left-hand side of this equation is an $n \times n$ matrix whose rows are linear combinations of the rows of $\operatorname{adj}(\mathbf{M})$. Because $\mathbf{M} \neq 0$ at least one of these linear combinations is non-trivial.

Alternative solution to (b).

It follows from the equation $\mathbf{M}\operatorname{adj}(\mathbf{M}) = \mathbf{0}$ that the columns of $\operatorname{adj}(\mathbf{M})$ are contained in the null space of $\mathbf{M}$.

Thus the column space of $\operatorname{adj}(\mathbf{M})$ is contained in the null space of $\mathbf{M}$.

Thus:

$$\operatorname{rank}(\operatorname{adj}(\mathbf{M})) \leq \operatorname{nullity}(\mathbf{M}) < n.$$

The last inequality follows from the rank-nullity theorem and the fact that $\mathbf{M}$ is not equal to zero.

Thus the dimension of the row space of $\operatorname{adj}(\mathbf{M})$ is less than n.

Thus the rows are linearly dependent.

□

Problem 9: (Similar to a hard quiz problem from 2018.)

Consider the following list S of vectors in $\mathbb{R}^5$.

$$S = \{(1,1,1,1,1),(1,0,1,0,1),(2,1,0,-1,1)\}.$$

Give equations on the parameters a,b,c,d,e which imply that the associated vector $v = (a,b,c,d,e)$ has the property that when you add it to the list S then the resulting list is linearly independent.

Solution

We want to know for which (a,b,c,d,e) is the following system inconsistent:

$$c_1(1,1,1,1,1)+c_2(1,0,1,0,1)+c_3(2,1,0,-1,1)=(a,b,c,d,e).$$

The corresponding augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 1 & 0 & 1 & b \\ 1 & 1 & 0 & c \\ 1 & 0 & -1 & d \\ 1 & 1 & 1 & e \end{array}\right].$$

Now do Gaussian elimination:

$$\to \left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & -1 & -1 & b-a \\ 0 & 0 & -2 & c-a \\ 0 & -1 & -3 & d-a \\ 0 & 0 & -1 & e-a \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & -1 & -1 & b-a \\ 0 & 0 & -2 & c-a \\ 0 & 0 & -2 & (d-a)-(b-a) \\ 0 & 0 & -1 & e-a \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & -1 & -1 & b-a \\ 0 & 0 & -1 & e-a \\ 0 & 0 & -2 & c-a \\ 0 & 0 & -2 & d-b \end{array}\right]$$

Continuing:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & a \\ 0 & -1 & -1 & b-a \\ 0 & 0 & -1 & e-a \\ 0 & 0 & 0 & (c-a)-2(e-a) \\ 0 & 0 & 0 & (d-b)-2(e-a) \end{array}\right]$$

This system will be inconsistent if

$$\text{either } a+c-2e \neq 0 \ \text{ or } \ 2a-b+d-2e \neq 0.$$

Note that your answer may look different depending on what steps you took in Gaussian elimination. If you got a different answer maybe see if you have obtained an equivalent system.

$\square$

Problem 10:

Assume that the following list of vectors in some $\mathbb{R}^m$ is linearly independent

$$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1}, \mathbf{v}_n.$$

Show that this implies that the following list

$$\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_2 - \mathbf{v}_3, \ldots, \mathbf{v}_{n-1} - \mathbf{v}_n, \mathbf{v}_n$$

is also linearly independent.

Solution

We know that the only solution to the equation

$$c_1\mathbf{v}_1 + \ldots + c_n\mathbf{v}_n = \mathbf{0}$$

is $c_1 = c_2 = \ldots = c)n = 0$.

We need to use this fact to determine the possible solutions to the equation

$$d_1(\mathbf{v}_1 - \mathbf{v}_2) + d_2(\mathbf{v}_2 - \mathbf{v}_3) + \ldots + d_{n-1}(\mathbf{v}_{n-1} - \mathbf{v}_n) + d_n\mathbf{v}_n = \mathbf{0}.$$

Rearranging this equation to group the $\mathbf{v}_i$ terms we get:

$$d_1\mathbf{v}_1 + (d_2 - d_1)\mathbf{v}_2 + (d_3 - d_2)\mathbf{v}_3 + \ldots + (d_n - d_{n-1})\mathbf{v}_n = \mathbf{0}.$$

Because we know the $\mathbf{v}_i$ are linearly independent we deduce the system of equations

$$\begin{aligned} d_1 &= 0 \\ d_2 - d_1 &= 0 \\ d_3 - d_2 &= 0 \\ &\vdots \\ d_n - d_{n-1} &= 0 \end{aligned}$$

These equations imply

$$d_1 = d_2 = \ldots = d_n = 0.$$

Thus the given list of vectors is linearly independent.

$\square$

Problem 11:

Let $\mathbf{v_1}, \ldots, \mathbf{v_p}$ be a collection of p vectors in $\mathbb{R}^m$ and let $\mathbf{P}$ be an invertible $m \times m$ matrix.

1. Show that $\mathbf{v_1}, \ldots, \mathbf{v_p}$ are linearly independent if and only if $\mathbf{Pv_1}, \ldots, \mathbf{Pv_p}$ are linearly independent.
2. Let V be a subspace of $\mathbb{R}^m$ that contains the vectors $\mathbf{v_1}, \ldots, \mathbf{v_p}$ and define W to be the subspace of $\mathbb{R}^m$ given by
$$W = \{\mathbf{Pv} \mid \mathbf{v} \in V\}.$$
Show that $\mathbf{v_1}, \ldots, \mathbf{v_p}$ span V if and only if $\mathbf{Pv_1}, \ldots, \mathbf{Pv_p}$ span W.

Solutions

Solution to (i): Since $\mathbf{P}$ is invertible we have, for any $x_1, \ldots, x_p \in \mathbb{R}$,

$$\begin{aligned} & x_1\mathbf{v_1} + x_2\mathbf{v_2} + \cdots + x_p\mathbf{v_p} = \mathbf{0} && (*) \\ \Leftrightarrow\quad & \mathbf{P}(x_1\mathbf{v_1} + x_2\mathbf{v_2} + \cdots + x_p\mathbf{v_p}) = \mathbf{0} \\ \Leftrightarrow\quad & x_1\mathbf{Pv_1} + x_2\mathbf{Pv_2} + \cdots + x_p\mathbf{Pv_p} = \mathbf{0} && (**) \end{aligned}$$

This shows that if $\mathbf{v_1}, \ldots, \mathbf{v_p}$ are linearly independent, i.e. $(*)$ is only satisfied when all x_j's are zero, then the same is true for $\mathbf{Pv_1}, \ldots, \mathbf{Pv_p}$ and vice-versa.

Solution to (ii): Note that for any vector $\mathbf{v}$ in V we have

$$\begin{aligned} & x_1\mathbf{v_1} + x_2\mathbf{v_2} + \cdots + x_p\mathbf{v_p} = \mathbf{v} && (*) \\ \Leftrightarrow\quad & \mathbf{P}(x_1\mathbf{v_1} + x_2\mathbf{v_2} + \cdots + x_p\mathbf{v_p}) = \mathbf{Pv} \\ \Leftrightarrow\quad & x_1\mathbf{Pv_1} + x_2\mathbf{Pv_2} + \cdots + x_p\mathbf{Pv_p} = \mathbf{Pv} && (**) \end{aligned}$$

We use this to show (a) if $\mathbf{v_1}, \ldots, \mathbf{v_p}$ span V then $\mathbf{Pv_1}, \ldots, \mathbf{Pv_p}$ span W, and (b) if $\mathbf{Pv_1}, \ldots, \mathbf{Pv_p}$ span W then $\mathbf{v_1}, \ldots, \mathbf{v_p}$ span V.

Proof of (a): We need to show that any $\mathbf{w} \in W$ can be written as a linear combination of $\mathbf{Pv_1}, \ldots, \mathbf{Pv_p}$. By definition of W there exists $\mathbf{v} \in V$ such that $\mathbf{w} = \mathbf{Pv}$. Since by assumption $\mathbf{v_1}, \ldots, \mathbf{v_p}$ span V we can write $\mathbf{v}$ as a linear combination of these vectors as in $(*)$ above. But then $(**)$ shows that $\mathbf{w} = \mathbf{Pv}$ can be written as a linear combination of $\mathbf{Pv_1}, \ldots, \mathbf{Pv_p}$ which was what we needed to show.

Proof of (b): We need to show that any $\mathbf{v} \in V$ can be written as a linear combination of $\mathbf{v_1}, \ldots, \mathbf{v_p}$. Note first that $\mathbf{Pv} \in W$. Since by assumption $\mathbf{Pv_1}, \ldots, \mathbf{Pv_p}$ span W we can write $\mathbf{Pv}$ as a linear combination of these vectors as in $(**)$ above. But then $(*)$ shows that $\mathbf{v}$ can be written as a linear combination of $\mathbf{v_1}, \ldots, \mathbf{v_p}$ which was what we needed to show.

$\square$